MUTASIM FOUAD SHOWMIK

Junior Software Engineer with hands-on experience in production-level software development, backend engineering, RESTful APIs, and Python-based machine learning applications.

+8801701106837 mutasim.fouad.showmik@gmail.com https://www.linkedin.com/in/fouad-showmik-34b9a9313/ Dhaka, Bangladesh

https://github.com/Fouad-Showmik?tab=repositories https://www.facebook.com/fouad.showmik/

PROFESSIONAL SUMMARY

Computer Science graduate and Junior Software Engineer with professional experience in developing, maintaining, and debugging production-level software applications. Proficient in Python, C#, Dart, and backend development, with hands-on experience building RESTful APIs, integrating databases, and implementing scalable application architectures. Experienced in ASP.NET Core, FastAPI, Entity Framework, Git-based collaborative workflows, and Docker. Additionally, have practical experience developing Python-based machine learning systems and integrating predictive models into application backends. Strong foundation in data structures, algorithms, object-oriented programming, and software engineering principles, with a keen interest in AI infrastructure and data-driven systems.

WORK EXPERIENCE

Junior Software Engineer (Full-Time) | Efficient Software Solutions Ltd.

August 2025 – September 2026

- Developed, maintained, and debugged production-level software applications using Flutter, Dart, and backend technologies.
- Integrated RESTful APIs, implemented asynchronous data handling, and managed data serialization/deserialization across application components.
- Collaborated with backend engineers on ASP.NET Core services using N-Layer Architecture, Repository Pattern, Entity Framework, and LINQ.
- Worked with C# backend components involving MediatR (CQRS), Strategy Pattern, and database operations.
- Implemented authentication and authorization workflows using JWT and integrated AWS S3 for secure file storage.
- Investigated, reproduced, troubleshot, and resolved application issues to improve system reliability and performance.
- Collaborated with backend developers, QA engineers, and UX designers through Agile development workflows using Git, GitHub/GitLab, and Jira.
- Coordinated asynchronous, message-driven workflows using RabbitMQ and MassTransit, and worked with Docker-based development environments.

TECHNICAL SKILLS

Programming Languages: Python, C#, Dart, Java, JavaScript

Backend Development: ASP.NET Core, RESTful APIs, Entity Framework Core, LINQ, MediatR, CQRS, Repository Pattern, JWT Authentication, Database-First Approach

Database & Data Management: MySQL, SQLite, ObjectBox, Elasticsearch, Redis, Relational Database Design, ORM, Data Modeling, Database Querying

Software Engineering & Architecture: Object-Oriented Programming (OOP), Data Structures & Algorithms, N-Layer Architecture, Clean Architecture, SOLID Principles, Design Patterns, Asynchronous Programming, Error Handling, Debugging

Testing & Quality Assurance: Code Review Fundamentals, API Testing, Input Validation, Troubleshooting

AI & Machine Learning: TensorFlow, Keras, Scikit-learn, Model Training & Evaluation, Feature Engineering, Missing Value Imputation, Hyperparameter Tuning, SHAP, LIME

AI-Assisted Development: LLM-Assisted Development Workflows, AI-Assisted Code Generation, Debugging & Error Analysis, Code Refactoring, Technical Documentation Assistance.

Tools & Platforms: Git, GitHub, GitLab, Docker, Jira, AWS S3, Firebase Crashlytics

Mobile Development: Flutter, Dart, Riverpod, KivyMD

Frontend & Web Development: HTML, CSS, JavaScript

Data Serialization & Communication: JSON, JSON Serialization/Deserialization, API Request/Response Handling

RECENT PROJECTS

- **FIFA 2026 Live Score Tracker** — Developed a real-time FIFA 2026 match tracking app in Flutter, featuring live score polling, team search, match filters, and stadium details. Integrated Riverpod for state management and Dio with custom interceptors for typed error handling and network-aware error states. Implemented parallel API fetching, shimmer loading, and a responsive match detail view with live goal scorers. Gained **100+** downloads on GitHub.
  https://github.com/Fouad-Showmik/Fifa_2026_Live_Score_Update_Native_App/tree/main
- **Alopecia Factors Detector** — Engineered a predictive backend system using Python and FastAPI to serve machine learning models that analyze medical and lifestyle risk factors. Alongside the backend, developed a cross-platform mobile application client utilizing Flutter and Dart to capture user health inputs and display real-time, insight-driven diagnostics. Architected secure and optimized HTTPS endpoints to manage full-stack data serialization, safely transferring user data from the Dart application layer to the predictive Python backend. Focused on building a highly reliable, low-latency API architecture to seamlessly integrate middleware services and deliver real-time model predictions directly to the client-side Flutter application.
  https://github.com/Fouad-Showmik/Alopecia-Hair-Loss-Factors-Detector
- **Library Management System** — Designed and developed a production-ready RESTful Library Management System following **N-Layer Architecture** with **Repository Pattern** and **MediatR CQRS**. Implemented **JWT Authentication** with role-based authorization (Admin/Member) using ASP.NET Core Identity. Integrated **RabbitMQ via MassTransit** for asynchronous event-driven notifications (book borrow/return confirmations) and **Elasticsearch** for high-performance full-text book search with fuzzy matching. Applied **Strategy Pattern** for dynamic late fee calculation based on member types. Containerized infrastructure services using **Docker Compose**.
  https://github.com/Fouad-Showmik/Library_Management_System_ASP_.NET_Core
- **Freight Rate Prediction:** I developed an end-to-end freight rate prediction project using Python and machine learning. The project involved exploratory data analysis, data cleaning, feature engineering, and model evaluation using time-based validation. I compared LightGBM and CatBoost against a baseline model and selected CatBoost based on its performance. Finally, I generated predictions for the validation dataset and a December scenario, with a focus on accuracy, reproducibility, and clear documentation.
  https://github.com/Fouad-Showmik/Freight-Load-Prediction-Rate_Machine-Learning_Assessment

EDUCATION

• **Secondary Education:** Comilla Zilla School, Comilla, Session (2016-17), Group: Science SSC (GPA-5.00)

• **Higher Secondary Education** Comilla Victoria Government College, Comilla, Session (2018-19), Group: Science HSC (GPA-5.00)

• **Undergraduate Education:** Brac University (BSC) Bachelor of Science in Computer Science and Engineering CGPA: 3.50 (4.00) | Graduated: December. 2024

RESEARCH & PUBLICATIONS

• Hybrid ML-SLM RAG System for Large Technical PDFs (IEEE Conference Publication)
https://ieeexplore.ieee.org/document/11118759 Contributed to research investigating a hybrid machine learning and Small Language Model (SLM) based Retrieval-Augmented Generation (RAG) system for large technical PDF documents.

• Advanced Machine Learning Techniques for Personalized Alopecia Areata Intervention Modeling (Undergraduate Thesis, BRAC University) https://drive.google.com/file/d/1qUUGnaktV9UsRe8KDqDDYGzrBwPq5RVa/view?usp=sharing Conducted machine learning research involving multiple datasets, predictive modeling, explainable AI, feature engineering, dataset integration, and a hybrid CNN-autoencoder architecture.

LANGUAGE

• Bangla (Native) • English (Professional)